

A nine-vertex counterexample to the crystal duality criterion

Source 00000003849: local formalization package

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Claim addressed. The source asserts that a crystal is isomorphic to its arrow-reversal dual if and only if its weight set is centrally symmetric, together with a polynomial-time verification claim. The finite counterexample below refutes the sufficient direction. Both source languages say weight *set*, with no connectedness or irreducibility restriction. No complexity theorem is established here.

The root datum and crystal convention

Use the type- A_2 weight lattice $P = \mathbb{Z}^2$ in fundamental-weight coordinates. The simple roots and their coroot pairings are

$$\alpha_1 = (2, -1), \quad \alpha_2 = (-1, 2), \quad \langle (a, b), \alpha_1^\vee \rangle = a, \quad \langle (a, b), \alpha_2^\vee \rangle = b.$$

Thus the Cartan matrix has diagonal entries 2 and off-diagonal entries -1 . The six roots are $\pm\alpha_1$, $\pm\alpha_2$ and $\pm(\alpha_1 + \alpha_2)$, with corresponding coroots $\pm(1, 0)$, $\pm(0, 1)$ and $\pm(1, 1)$. The pairing of the weight and coweight lattices is the standard dot product. Lean checks that corresponding roots and coroots pair to 2, are distinct, and are closed under the corresponding simultaneous reflections. In particular, the simple reflections are

$$s_1(a, b) = (-a, a + b), \quad s_2(a, b) = (a + b, -b).$$

We use the finite-type Kashiwara crystal convention. A crystal consists of a weight map, partial operators e_i, f_i and integer string functions ε_i, φ_i . The partial inverse relation and lowering relations are

$$\begin{aligned} f_i(x) = y &\iff e_i(y) = x, \\ \text{wt}(y) &= \text{wt}(x) - \alpha_i, & \varepsilon_i(y) &= \varepsilon_i(x) + 1, & \varphi_i(y) &= \varphi_i(x) - 1, \end{aligned}$$

with the corresponding raising relations and $\varphi_i(x) = \varepsilon_i(x) + \langle \text{wt}(x), \alpha_i^\vee \rangle$. The absent vertex is Lean's `Option.none`. These are a specialization of the usual abstract axioms to finite string values, not assumed admissibility. All axioms are proved for the constructed crystal. Exact nonzero operator string lengths are also proved, so the example is seminormal.

The actual finite crystal

Let S have three vertices p_0, p_1, p_2 , weights

$$(1, 0), \quad (-1, 1), \quad (0, -1),$$

and arrows $p_0 \xrightarrow{1} p_1 \xrightarrow{2} p_2$. For each color, the raising operator is the reverse partial arrow; its string functions are 0 or 1 according to whether the corresponding arrow exists. This is the

standard fundamental A_2 crystal $B(\omega_1)$. For an explicit weight correspondence, the standard GL_3 weights e_1, e_2, e_3 map under

$$(u_1, u_2, u_3) \mapsto (u_1 - u_2, u_2 - u_3)$$

to the three listed weights. The same map sends the two GL_3 simple roots to α_1, α_2 , and kills the central direction. These identities are also proved in Lean. The name of the component is therefore supported by its precise graph and weight correspondence; quantum-group representation theory is not needed as an unproved premise.

For any crystal C , its dual $C^\vee$ has

$$\text{wt}_C^\vee(v) = -\text{wt}_C(v), \quad e_i^\vee = f_i^C, \quad f_i^\vee = e_i^C, \quad \varepsilon_i^\vee = \varphi_i^C, \quad \varphi_i^\vee = \varepsilon_i^C.$$

The generic Lean construction proves all crystal axioms for this dual. The dual of S has highest weight $(0, 1)$ and is the standard dual component $B(\omega_2)$.

Set $B = S \sqcup S \sqcup S^\vee$. Its vertex set is $\{0, 1, 2\} \times \{0, 1, 2\}$. The first coordinate selects the component. All operators preserve that coordinate. The two first components use S ; the last uses the actual constructed dual. Lean proves the component correspondence for weights, both operators and both string functions.

Component	Weights at positions 0, 1, 2	Arrows
0	$(1, 0), (-1, 1), (0, -1)$	$0 \xrightarrow{1} 1 \xrightarrow{2} 2$
1	$(1, 0), (-1, 1), (0, -1)$	$0 \xrightarrow{1} 1 \xrightarrow{2} 2$
2	$(-1, 0), (1, -1), (0, 1)$	$2 \xrightarrow{2} 1 \xrightarrow{1} 0$

Symmetry and non-isomorphism

The weight set of B is

$$\{\pm(1, 0), \pm(-1, 1), \pm(0, -1)\}.$$

It is invariant under negation, hence centrally symmetric with centre zero. Lean states this for the actual set $\text{range}(\text{wt}_B)$, using the definition $\exists c \in P \forall w \in P : w \in X \iff 2c - w \in X$.

Lemma 1. *No bijection $g : B \rightarrow B$ satisfies $\text{wt}_B(g(v)) = -\text{wt}_B(v)$ for every vertex.*

Proof. The distinct vertices $(0, 0)$ and $(1, 0)$ both have weight $(1, 0)$. The only vertex of weight $(-1, 0)$ is $(2, 0)$. The required condition would send both distinct vertices to $(2, 0)$, contradicting injectivity. $\square$

Theorem 1. *The admissible nine-vertex crystal B has centrally symmetric weight set, but $B^\vee$ is not isomorphic to B as a crystal.*

Proof. A crystal isomorphism is an actual bijection preserving weights, both crystal operators and both string functions. If $g : B^\vee \rightarrow B$ were such an isomorphism, weight preservation and the definition of the dual would give $\text{wt}_B(g(v)) = \text{wt}_B^\vee(v) = -\text{wt}_B(v)$, contrary to the lemma. The reverse isomorphism direction is excluded by the same proved bridge. $\square$

This refutes the universal sufficient direction and the source's iff already within nonempty finite-type A_2 crystals. A counterexample in that subclass also refutes an assertion made for unqualified crystals. It does not replace the conjecture by a connected-crystal assertion or a multiset assertion. The algorithmic conjunct need not be resolved to refute a conjunction whose iff conjunct is false.

Formal artifact correspondence

The Main module imports the formal crystal source. In the namespace `CrystalCounterexample`, the key declarations are listed below.

```
package/CrystalCounterexample.lean
original_sufficient_direction_false
original_iff_false
Crystal; Crystal.dual; Crystal.Iso
Witness.crystal; Witness.seminormal
witness_centrallySymmetric; Witness.not_selfDual
```

Preparation compiles use official Lean 4.33.0 and fixed mathlib revision

```
db584cd6d46c92f209a44c0f1c829460d327499d
```

Printed transitive axiom dependencies are limited to the standard three:

```
propext; Classical.choice; Quot.sound
```

Preparation evidence is not manager acceptance: independent source-semantic review and a clean deterministic gate remain separate steps. No polynomial-time result is claimed.

References

- [1] E. Marberg, *Combinatorics of crystal bases*, MATH 6150I, Lecture 5, HKUST, Spring 2020. Definitions 2.1 and 3.4 and Section 4 supply the standard crystal, dual and isomorphism conventions. https://www.math.hkust.edu.hk/~emarberg/teaching/2020/Math6150I/lectures/05_Math6150I_Spring2020.pdf